

MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 85 | Date: 14-03-2026

Total Questions: 15 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQS

1. Which of the following best describes the intent of the Adapter pattern? [2 Marks]
 - a) Decouple an abstraction from its implementation.
 - b) Attach additional responsibilities to an object dynamically.
 - c) Convert the interface of a class into another interface clients expect.
 - d) Provide a unified interface to a set of interfaces in a subsystem.

2. In the Decorator pattern, how does it primarily offer an alternative to extending functionality compared to static inheritance? [2 Marks]
 - a) By decoupling an abstraction from its implementation.
 - b) By composing objects into tree structures.
 - c) By attaching additional responsibilities to an object dynamically.
 - d) By providing a surrogate or placeholder for another object.

3. The Flyweight pattern separates which two types of state to support a large number of objects efficiently? [2 Marks]
 - a) Public and private state.
 - b) Intrinsic and extrinsic state.
 - c) Global and local state.
 - d) Static and dynamic state.

4. What is the primary benefit of the Mediator pattern regarding object interaction? [2 Marks]
 - a) It simplifies aggregate interfaces by separating traversal logic.
 - b) It encapsulates a request as an object, supporting undoable operations.
 - c) It promotes loose coupling by keeping objects from referring to each other explicitly.
 - d) It allows an object to alter its behavior when its internal state changes.

5. In the Chain of Responsibility pattern, how is a request typically handled? [2 Marks]
 - a) A centralized object processes all requests.
 - b) The sender explicitly specifies the receiver.
 - c) The request is passed along a chain of handlers until one handles it.
 - d) All objects in the chain handle the request simultaneously.

SECTION A: SHORT ANSWERS

6. Explain how the Bridge pattern supports improved extensibility for both abstractions and implementations. [5 Marks]

7. Describe how the Composite pattern allows clients to treat individual objects and compositions of objects uniformly. **[5 Marks]**
8. How does the Facade pattern promote weak coupling between a subsystem and its clients? **[5 Marks]**
9. In the Memento pattern, what is the role of the Caretaker, and why is its interaction with the Memento object restricted? **[5 Marks]**
10. Explain how the Iterator pattern simplifies the aggregate interface. **[5 Marks]**

SECTION A: LONG ANSWERS

11. Describe the Adapter pattern. Include its intent, illustrate its two implementation approaches with their structures, and explain two scenarios where it is useful. **[10 Marks]**
12. Explain the Command pattern. Detail its intent, structure, and how it enables functionalities such as undo/redo and logging. Provide two benefits of using this pattern. **[10 Marks]**
13. Describe the State pattern. Explain its intent and structure, clarifying how an object appears to change its class based on internal state. Provide two scenarios for when to use it. **[10 Marks]**
14. Explain the Template Method pattern. Describe its intent and structure, discuss the "Hollywood Principle" in its context, and list two consequences of using it. **[10 Marks]**
15. Describe the Visitor pattern. Explain its intent and structure, focusing on how it allows defining new operations on an object structure without changing the classes of its elements. Provide two consequences of using this pattern. **[10 Marks]**

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